

Office simulation in packet tracer

Configuring DHCP Services on a Cisco 2911 Router Using Packet Tracer

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Objective

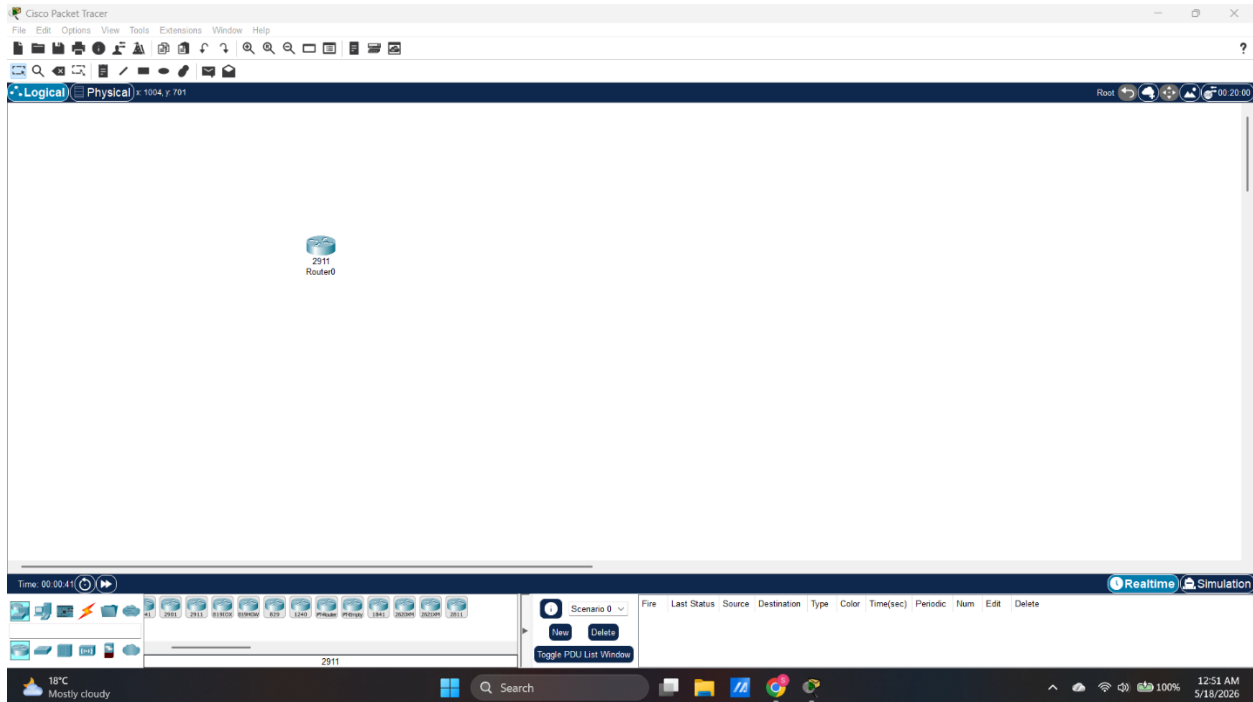
The objective of this lab is to configure a Cisco 2911 router as a DHCP server in Cisco Packet Tracer. The network simulates a small office environment where multiple PCs automatically receive IP addresses from the router. Connectivity between devices is verified using ping commands to ensure successful communication across the network.

Devices required

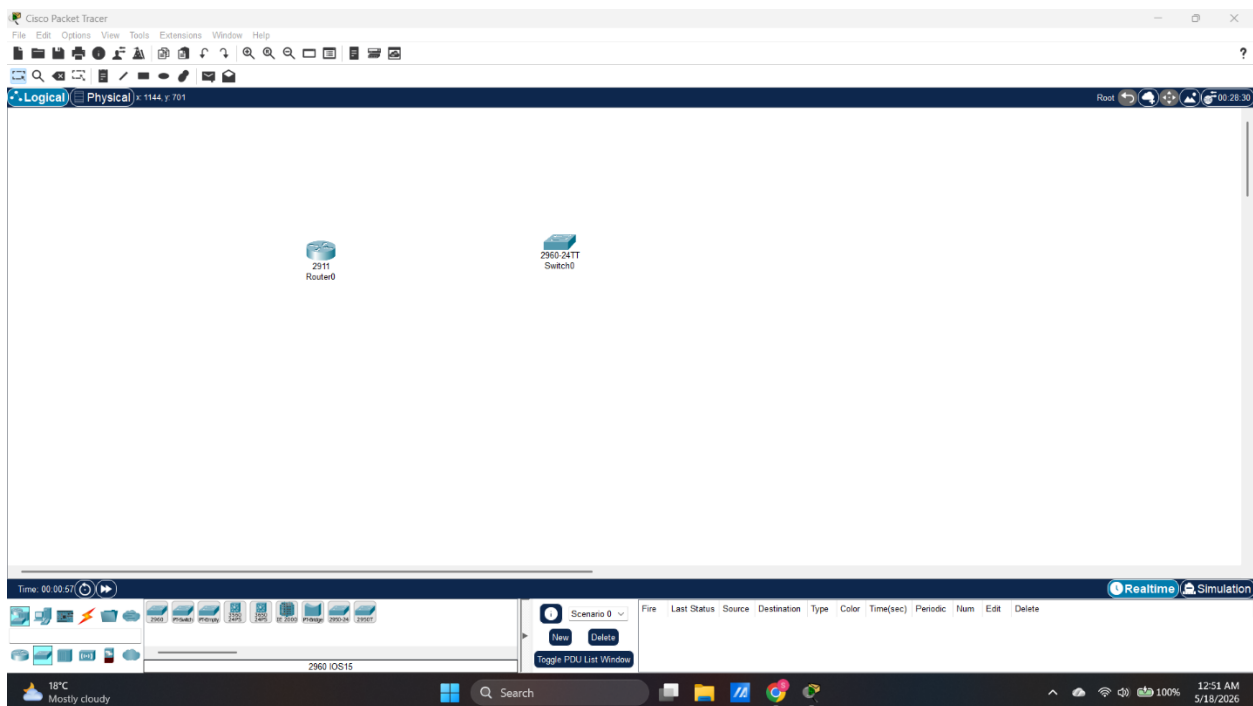
The following devices were used to build and simulate the network in packet tracer:

- Router (2911)
- Switch(2960-24TT)
- End devices -3 PC
- Cisco packet tracer

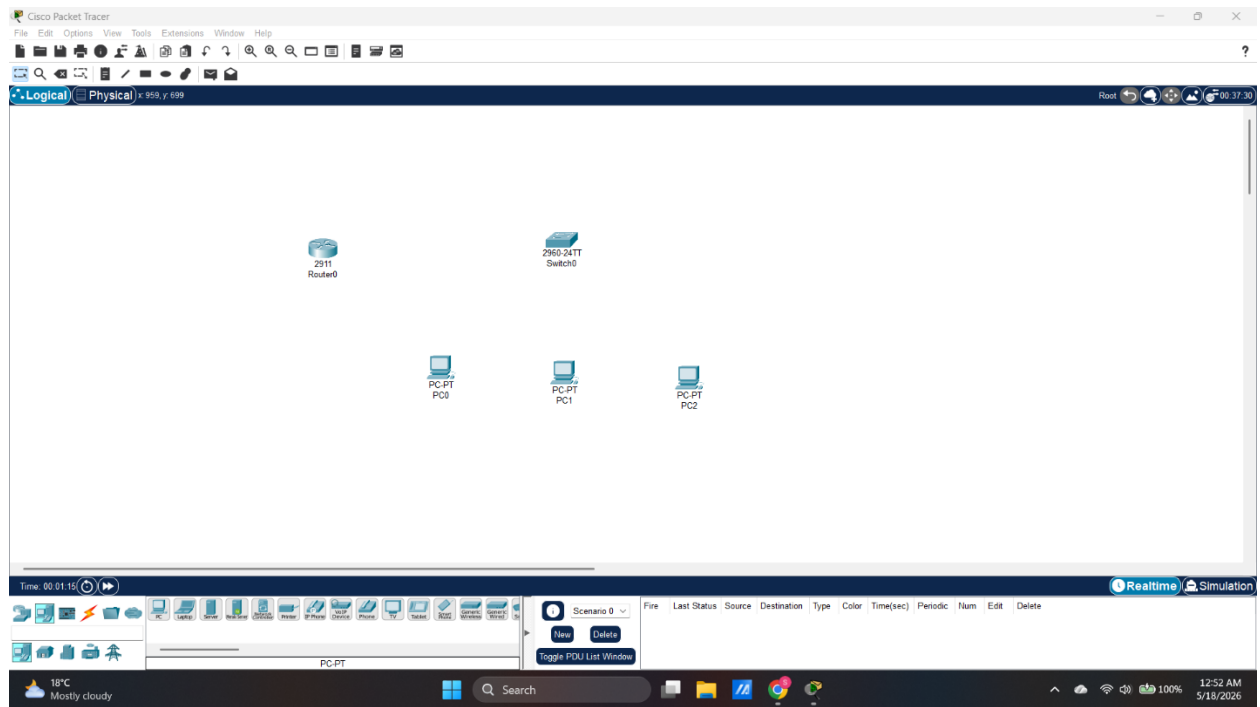
Devices needed for topology



First, take a router 2911.

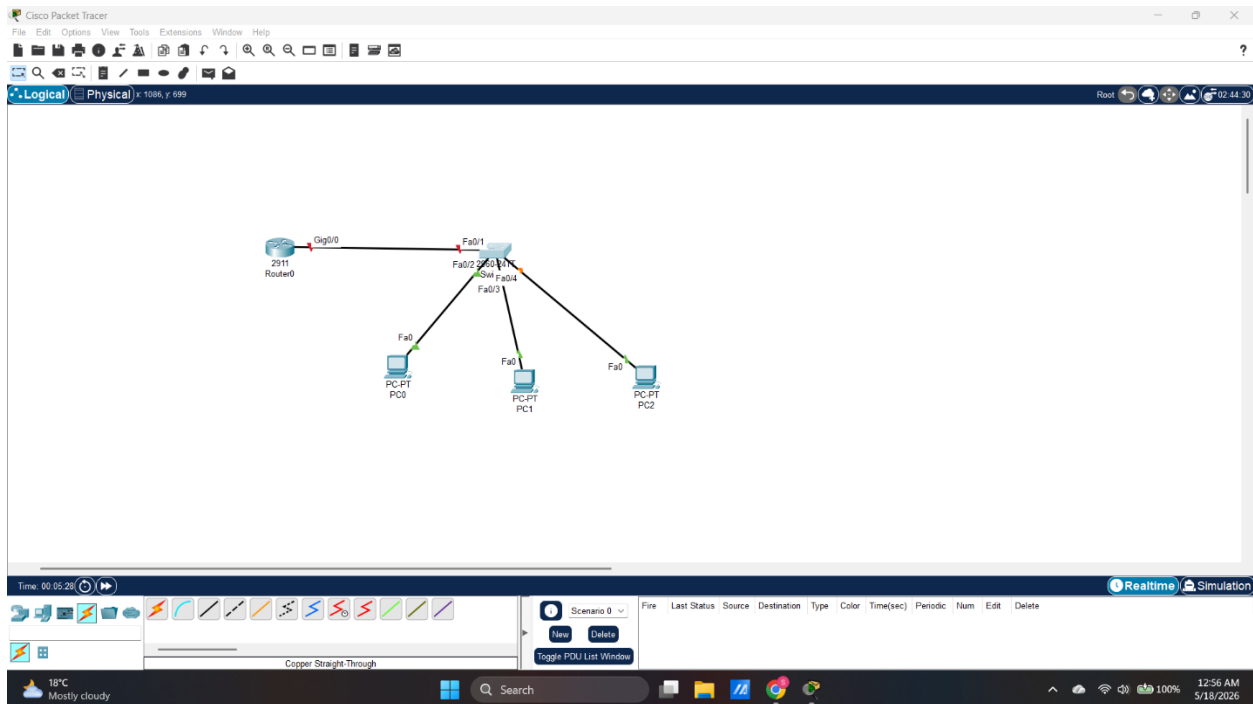


Take switch 2960-24TT.



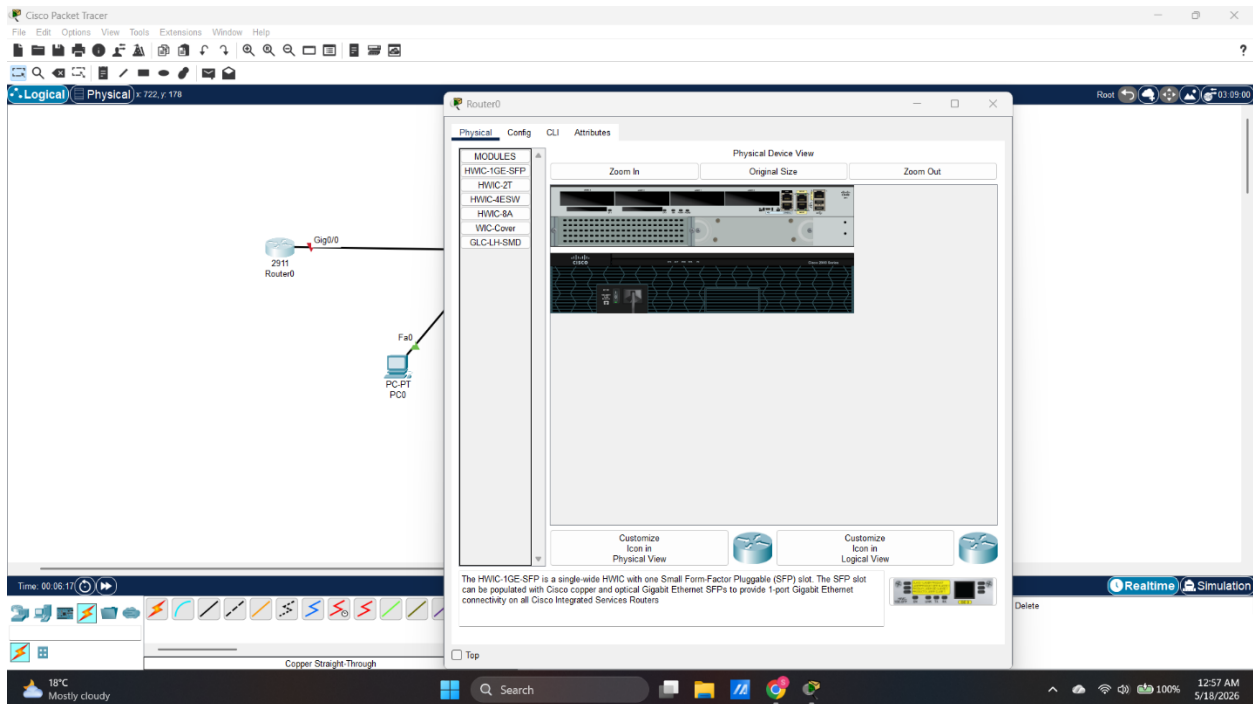
Add 3 PC.

Connecting all devices in topology

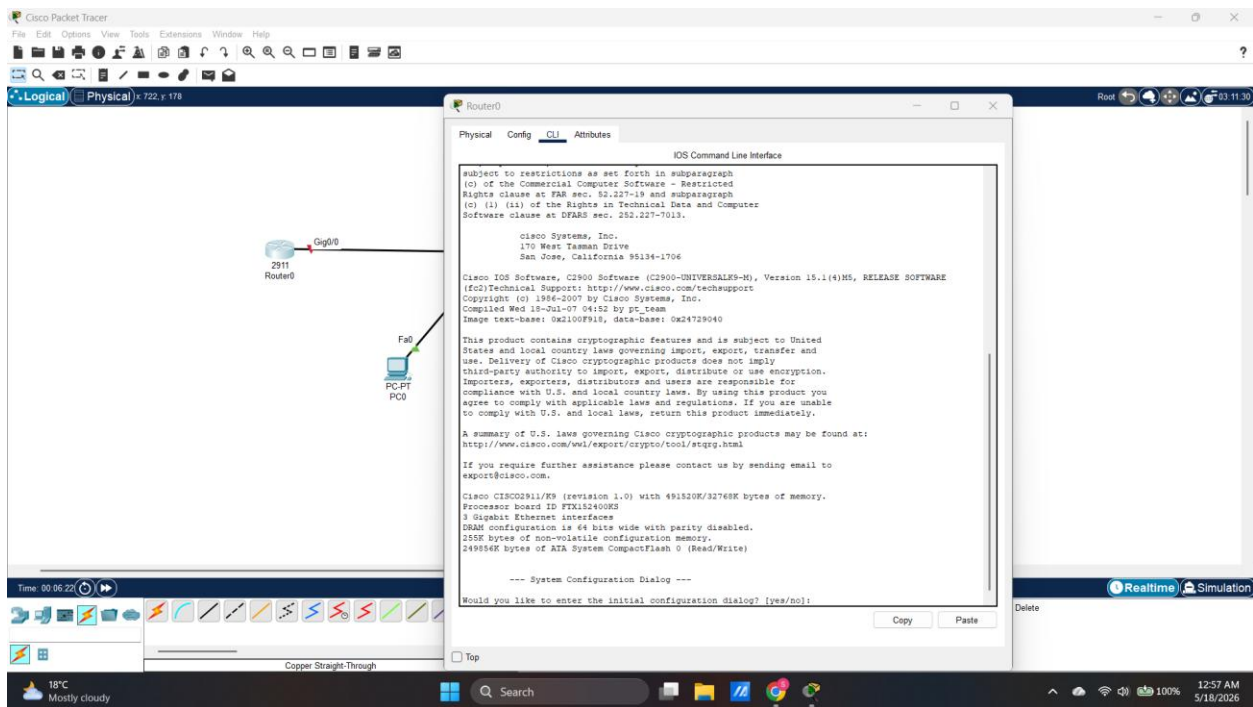


Now, use Copper Straight through cable and connect to the respective ports as shown in the given picture.

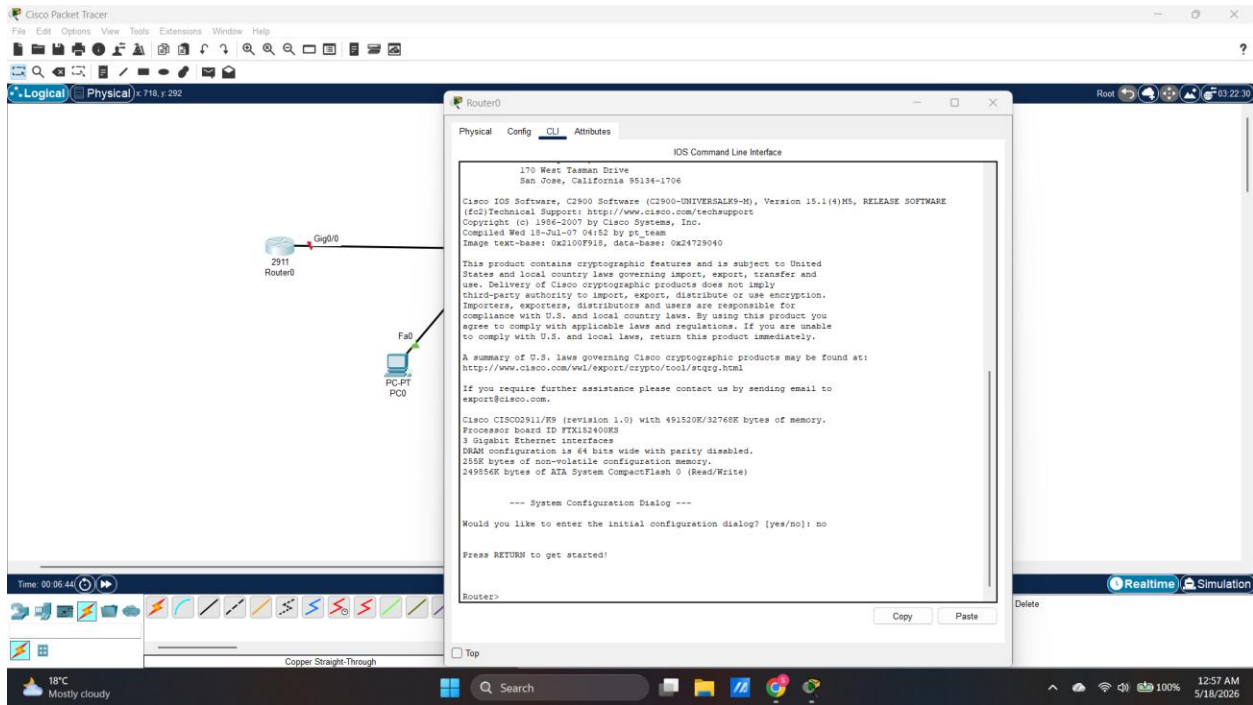
Router configuration



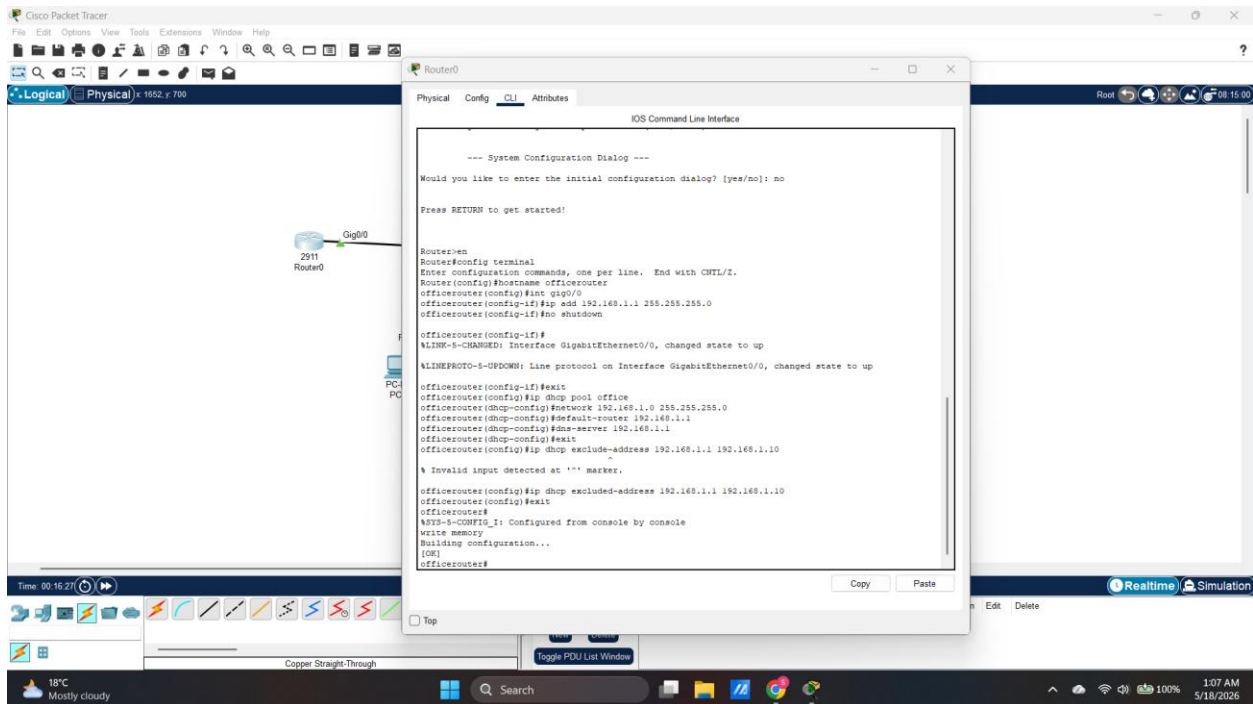
Next, open the router.



After that, go to CLI.

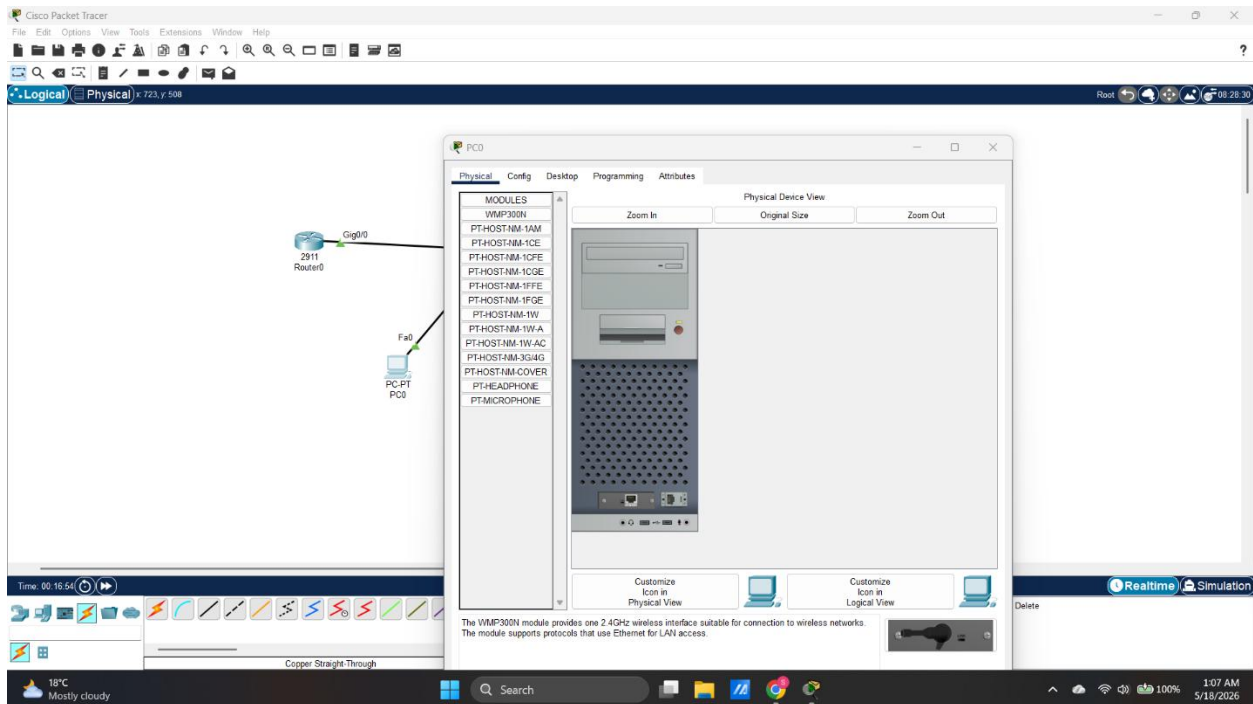


To go to system configuration dialog type “No”.

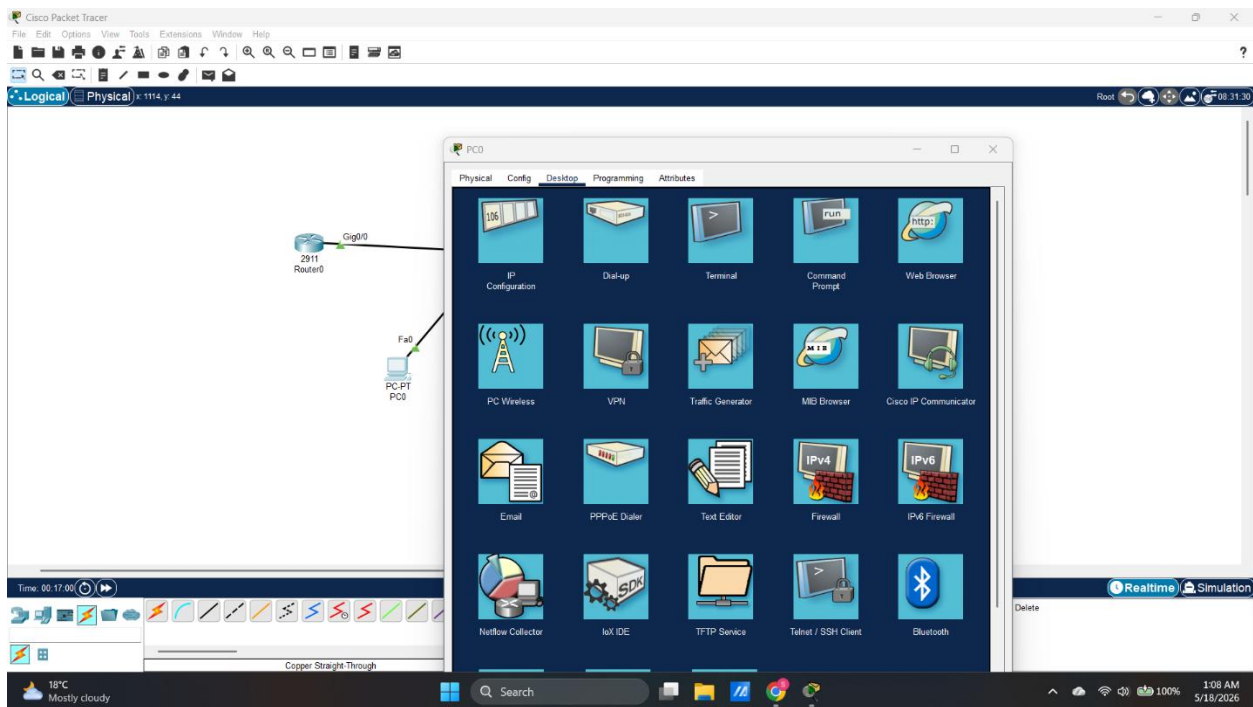


In CLI, use command to set the IP address for g0/0. Set address for DHCP and exclude IP address for routers.

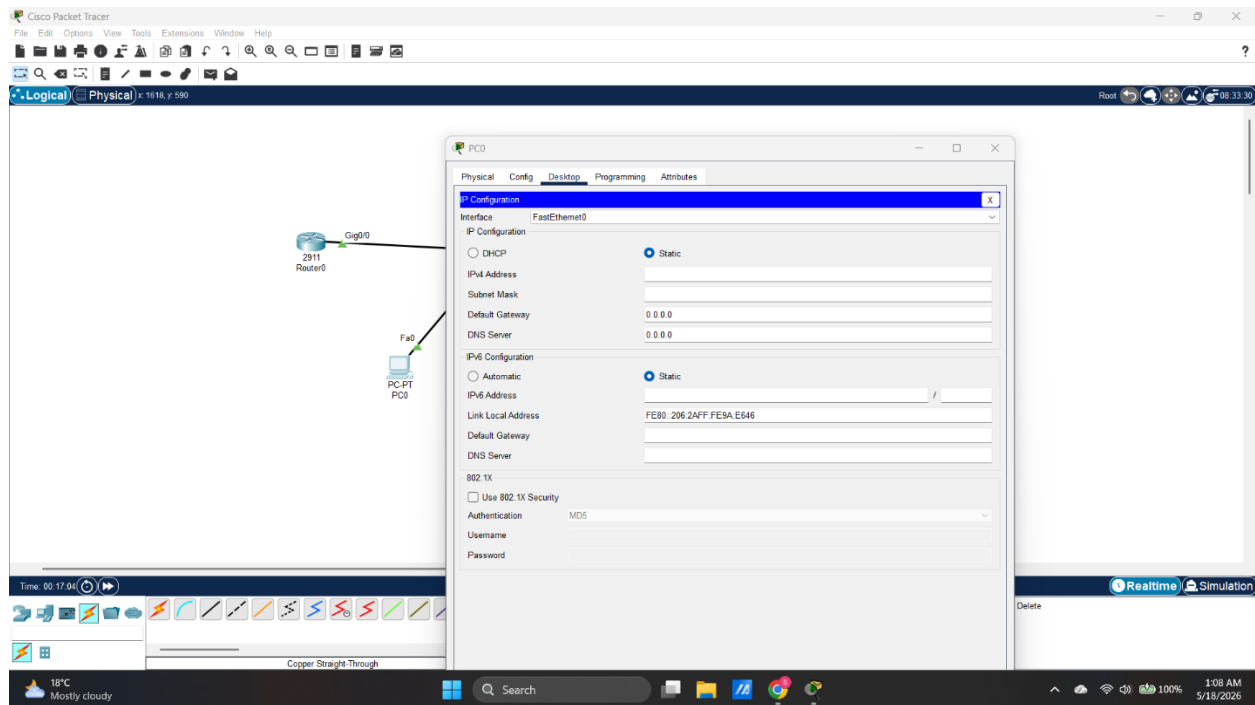
Changing static IP addressing to DHCP



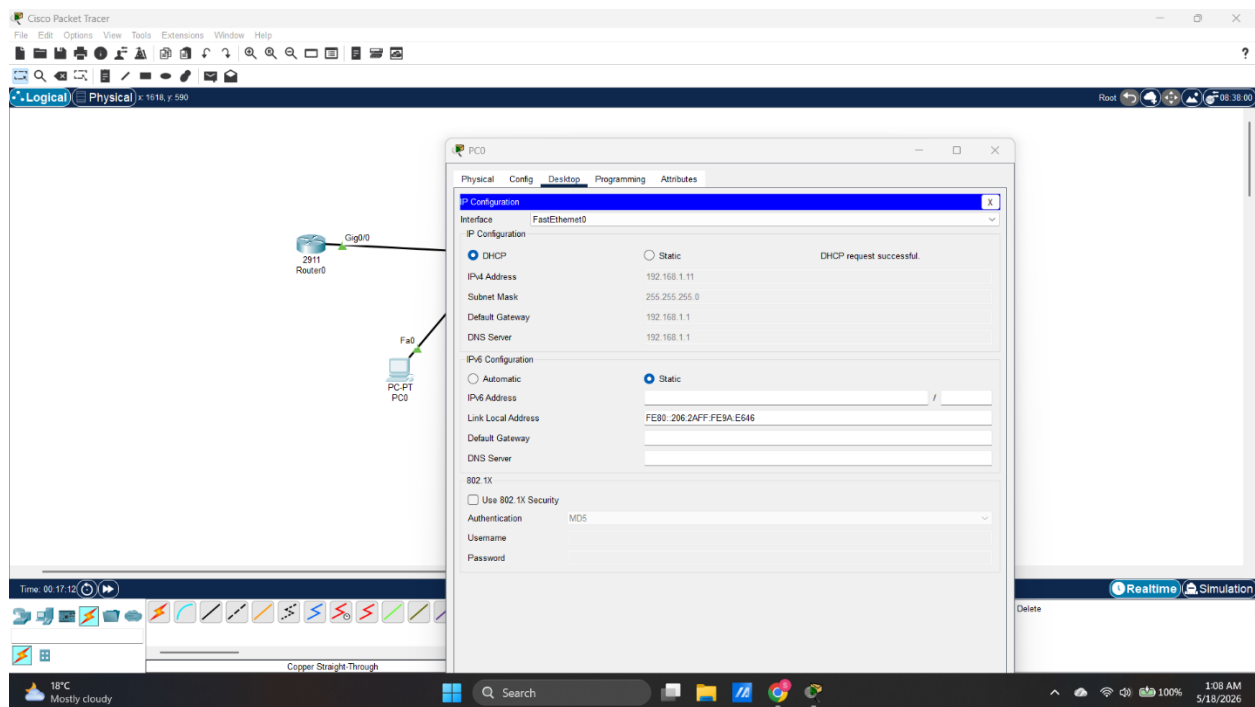
Then, go to PC0.



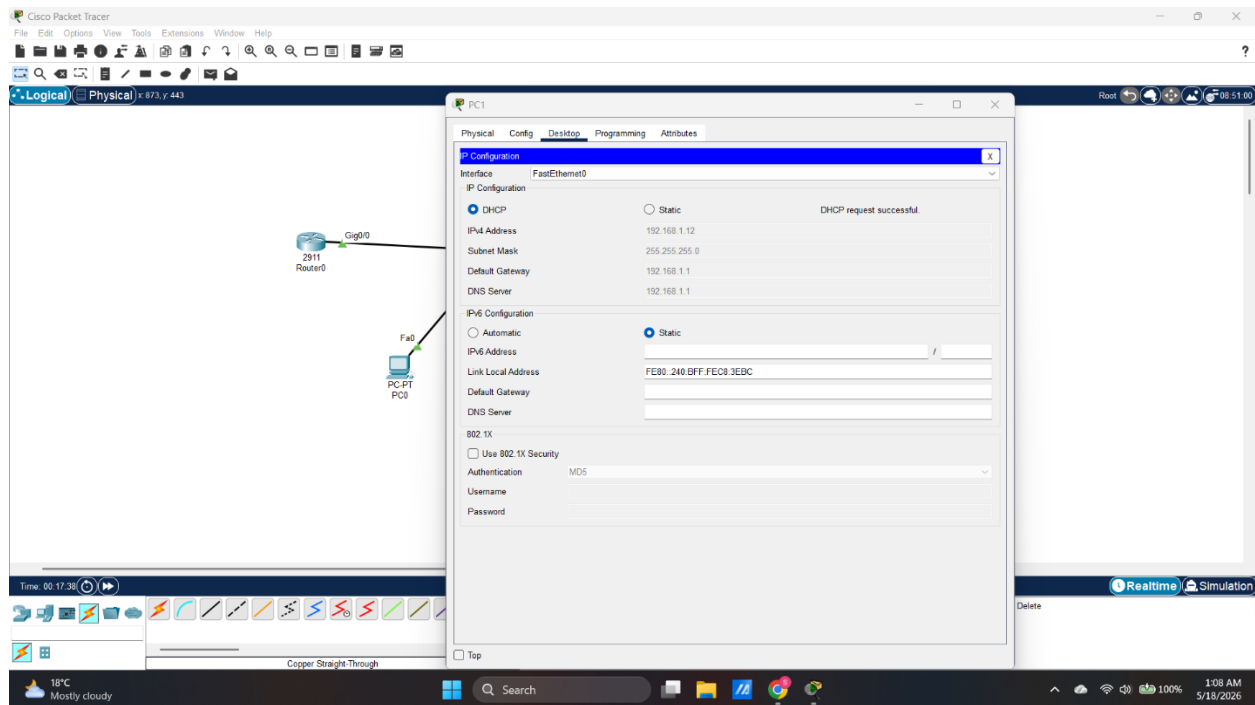
Click on desktop.



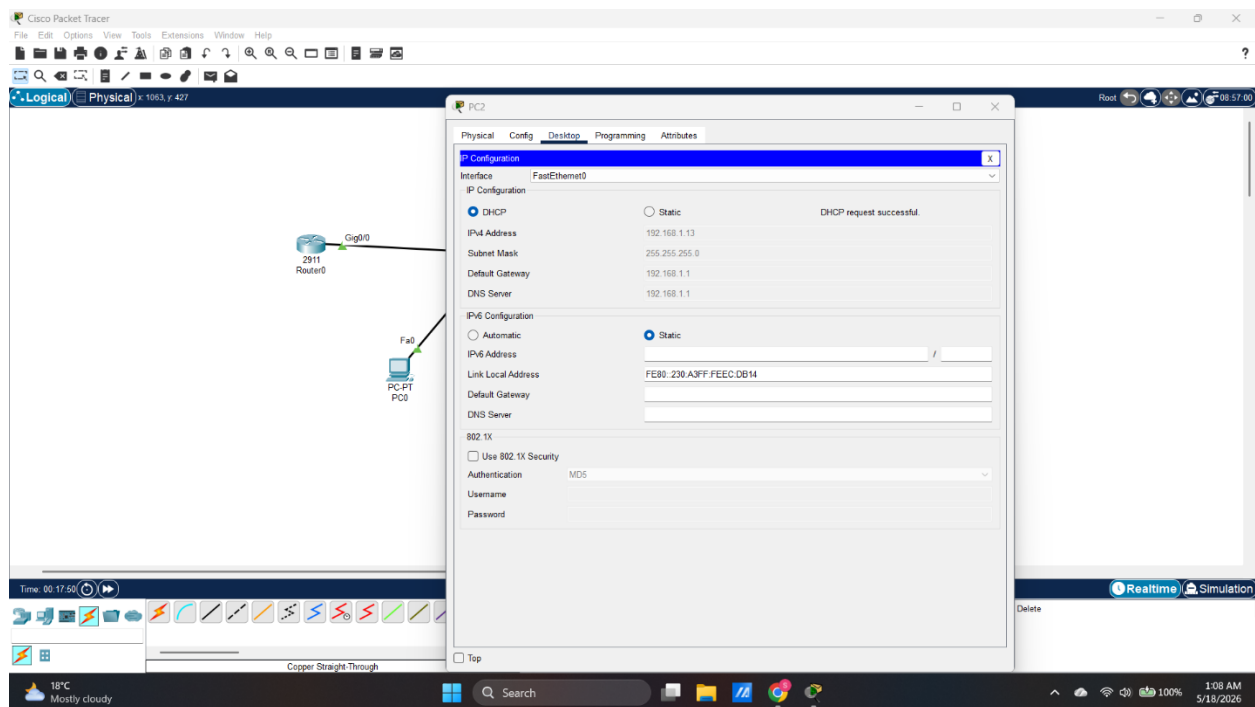
After that, open IP configuration.



Change fastEthernet interface to DHCP from static.

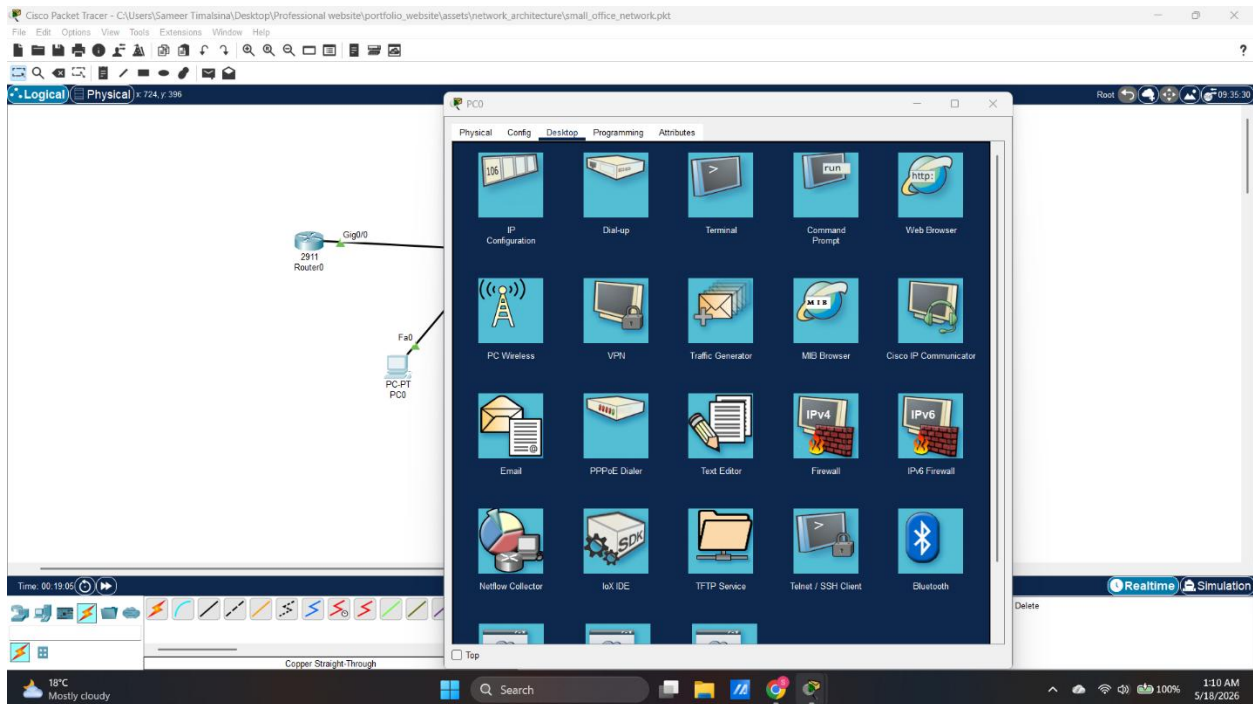


Similarly, for PC1 as well choose DHCP.

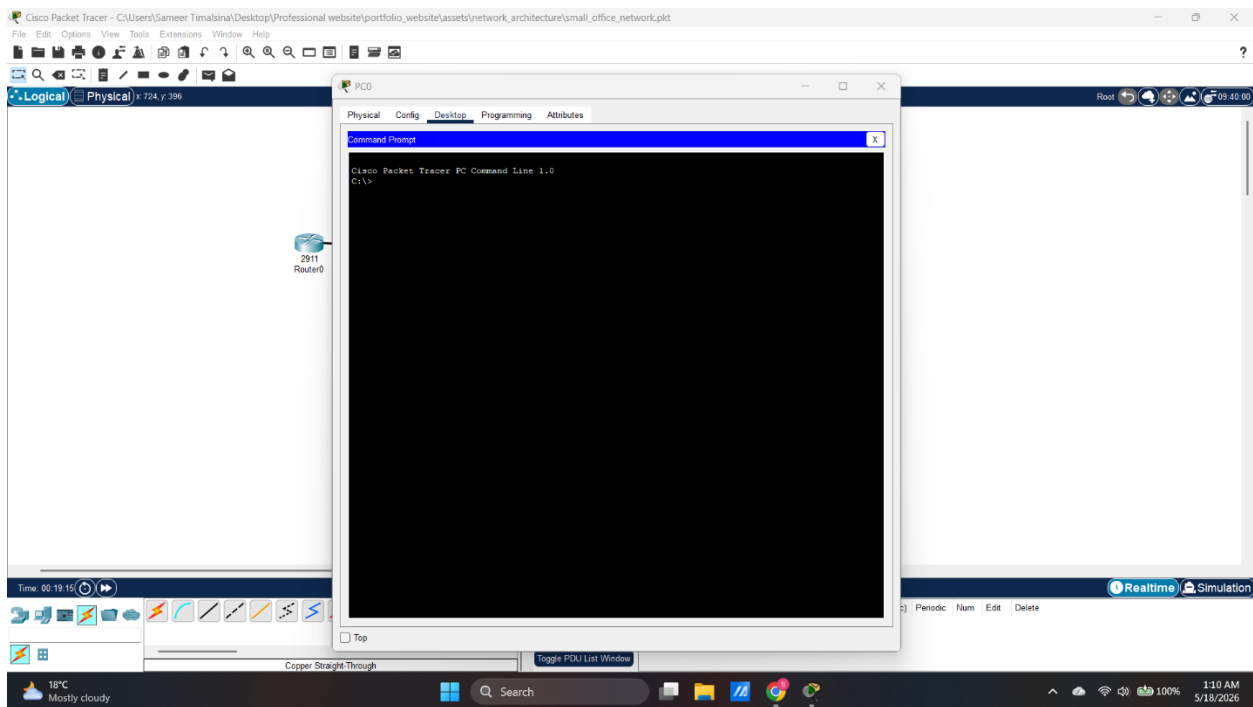


Do same for PC2 as well.

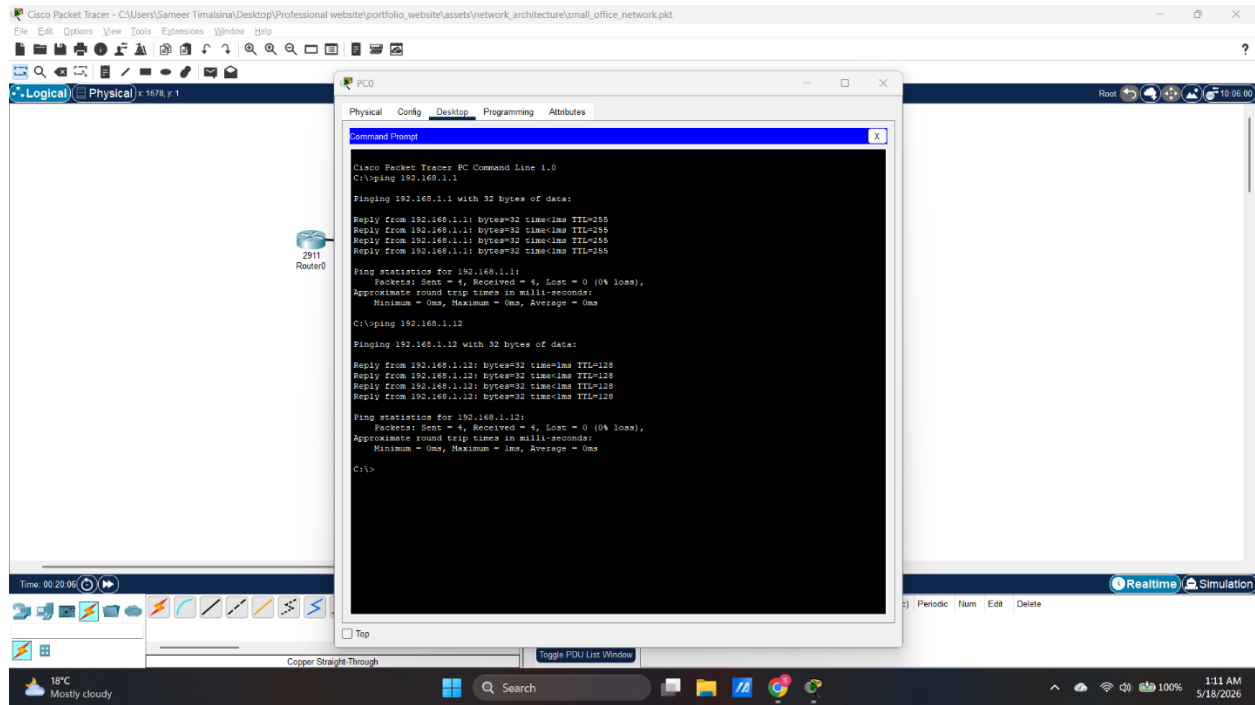
Verifying the connection



Again, go to PC0 desktop option.



Then open command prompt.



Here, ping both router and pc1. Ping command should be successful.

Configuration command

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname office-router
```

```
office-router(config)# interface gig0/0
```

```
office-router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
office-router(config-if)# no shutdown
```

```
office-router(config-if)# exit
```

```
office-router(config)# ip dhcp pool office
```

```
office-router(dhcp-config)# network 192.168.1.0 255.255.255.0
```

```
office-router(dhcp-config)# default-router 192.168.1.1
```

```
office-router(dhcp-config)# dns-server 8.8.8.8
```

```
office-router(dhcp-config)# exit
```

```
office-router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

```
office-router(config)# end
```

```
office-router# write memory
```

Conclusion

In this lab, a small office network was successfully designed and implemented using Cisco Packet Tracer. A Cisco 2911 router was configured as a DHCP server to automatically assign IP addresses to connected client devices. The network included a switch and multiple PCs connected in a star topology.

After configuration, all client devices were able to obtain IP addresses dynamically from the DHCP pool, and successful connectivity was verified using ping tests between devices. This confirms that the DHCP service is functioning correctly within the network.

This project helped reinforce the understanding of DHCP configuration, router setup, and basic network troubleshooting in a simulated environment.